

DD: BIL-CFG-01 v1.0 — BillableEvent Catalog

Developer implementation guide: DD_BIL-CFG-01-Billable_Event_Catalog-DEV_IMPL_GUIDE-v1.0.md.

1. Purpose

This rewritten DD is the developer-facing version of the module contract
DD_BIL-CFG-01-Billable_Event_Catalog-v1.0.md.

Verdict: ■ Updated | **Wave:** 1 | **Group I:** Billing | **Version:** v1.0 The BillableEvent catalog. Owns: the billable_event table (the entries that BIL-01 charges customers under), the billable_event_category sub-catalog (finance reporting classification), and the rules governing trigger types, applicability, signed-amount semantics, state callbacks, and operator scoping. BIL-01 reads this catalog at runtime to resolve which BillableEvents to charge when a workflow's bil01-emit-intent arrives. **What changed from v0.8:** - **Operator scoping** — operator_code column on both billable_event and billable_event_category. Each operator carries its own catalog. Per memory rule #16. - **Signed-amount semantics** — new amount_sign_policy column declaring whether the event's amount is always positive (charge), always negative (credit), or signed at runtime (computed by BIL-01 / billing-rules KJAR per-call). Used by MIGRATION_PRORATE which may be a charge or credit depending on source vs target Package price. - **Cycle-anchor support** — three new catalog rows added to the KE seed: OLD_CYCLE_CREDIT, FIRST_CYCLE_CHARGE, and a refined MIGRATION_PRORATE_DELTA with signed semantics. These cover the cycleAnchorPolicy = RESET_TO_X path used by UPGRADE/DOWNGRADE/MIGRATION. - **Section cleanup** — removed Module placement, Codebase, Foundation-specific configurations, Frontend, Gap & bridge, Migration approach, What's NEW in V3 sections per memory rule #26. The DD now specifies the catalog only.

The goal of this rewrite is to make the DD easier for a mid-level developer to implement without losing the detailed source contract.

2. Implementation Scope

This DD should be implemented as part of the owning SOPHIX capability, not as an isolated service unless the deployment architecture explicitly separates that capability.

Implement this DD with these rules:

- Use the owning module APIs/repositories for authoritative writes.
- Use approved internal APIs/client wrappers when reading another module's data.
- Do not query another module's database tables directly from business flow code.
- Treat worker names as logical Camunda external-task topics, not separate deployments.
- One worker application may handle multiple topics, but metrics, retries, and logs must remain visible per topic.
- Emit success events only after the authoritative state commit succeeds.
- Use outbox publishing for Kafka events.

3. Runtime Metadata

Item	Value
Source file	/Users/macbook/Downloads/Sophix_V3_BSS_DDs_MD_2026-05-27/05_billing/DD_BIL-CFG-01-Billable_Event_Catalog-v1.0.md
Version	v1.0
DD type	module contract
BPMN/process keys	Not explicitly defined
Drools packages	Not explicitly defined

4. Runtime Contracts To Implement

4.1 APIs

- GET /api/billing/billable-events/{id}

- GET /api/billing/billable-events?operatorCode=WIK&triggerType=SAGA_INTENT&triggerIntentCode=RELOCATION_INTENT&status=ACTIVE
- PATCH /api/billing/billable-events/{id}
- POST /api/billing/billable-events
- POST /api/billing/billable-events/{id}/activate
- POST /api/billing/billable-events/{id}/retire

4.2 Tables

- billable_event
- billable_event_category

For each table in this DD, implement the schema with:

- a short table comment explaining what the table is for;
- column comments or migration notes explaining each field;
- constraints and indexes from the DD;
- seed data where the DD provides operator-specific sample rows.

4.3 Worker Topics

- bil01-emit-intent
- billing-rules

Worker-topic guidance:

- A BPMN service task uses the topic.
- A worker application subscribes to the topic.
- The topic handler validates input variables, calls the owning module/API, writes outputs back to Camunda, and records metrics.
- Do not treat the topic string as a shell command or Kubernetes deployment name.

4.4 Events

- SubscriptionPaused
- SubscriptionTerminated

Event guidance:

- Write events through the transactional outbox.
- Include operation id, aggregate id, operator code, actor, and correlation data.
- Consumers should be able to rebuild downstream state without querying workflow internals.

5. Developer Implementation Checklist

1. Add or verify database migrations and seed rows.
2. Add or verify config rows for the operator/module.
3. Implement request/command DTOs and validation.
4. Implement idempotency and in-flight operation checks where the DD requires them.
5. Implement Drools fact mapping and package deployment where rules are used.
6. Implement BPMN process, service tasks, gateways, timers, and message catches where the DD is a flow.
7. Implement worker-topic handlers using owning module APIs.
8. Implement compensation paths and manual recovery paths.
9. Emit success/rejected events through the outbox.

10. Add smoke tests for happy path, validation failure, dependency failure, and retry/idempotency.

6. Infrastructure And AWS Notes

Deploy this DD with the existing SOPHIX platform components unless the owning module requires isolation:

Concern	Recommendation
API/runtime	Run inside the owning module deployment or existing workflow API pods.
Workers	Consolidate low-volume topics into shared worker apps; keep per-topic metrics.
Database	Use the owning module PostgreSQL schema and migration pipeline.
Kafka	Use the foundation Kafka/MSK setup and transactional outbox.
Camunda	Deploy BPMN files through the workflow deployment pipeline.
Drools	Deploy KJAR/rule artifacts through the approved rules pipeline.
Secrets	Use the platform secret manager; on AWS prefer Secrets Manager/Parameter Store with IRSA.
Observability	Alert on stuck operations, failed workers, outbox backlog, and external dependency failures.

7. Original DD Detail Preserved

The following section preserves the detailed source contract for implementation and review.

Verdict: ■ Updated | **Wave:** 1 | **Group I:** Billing | **Version:** v1.0

The BillableEvent catalog. Owns: the billable_event table (the entries that BIL-01 charges customers under), the billable_event_category sub-catalog (finance reporting classification), and the rules governing trigger types, applicability, signed-amount semantics, state callbacks, and operator scoping. BIL-01 reads this catalog at runtime to resolve which BillableEvents to charge when a workflow's bil01-emit-intent arrives.

What changed from v0.8:

- **Operator scoping** — operator_code column on both billable_event and billable_event_category. Each operator carries its own catalog. Per memory rule #16.

- **Signed-amount semantics** — new amount_sign_policy column declaring whether the event's amount is always positive (charge), always negative (credit), or signed at runtime (computed by BIL-01 / billing-rules KJAR per-call). Used by MIGRATION_PRORATE which may be a charge or credit depending on source vs target Package price.

- **Cycle-anchor support** — three new catalog rows added to the KE seed: OLD_CYCLE_CREDIT, FIRST_CYCLE_CHARGE, and a refined MIGRATION_PRORATE_DELTA with signed semantics. These cover the cycleAnchorPolicy = RESET_TO_X path used by UPGRADE/DOWNGRADE/MIGRATION.

- **Section cleanup** — removed Module placement, Codebase, Foundation-specific configurations, Frontend, Gap & bridge, Migration approach, What's NEW in V3 sections per memory rule #26. The DD now specifies the catalog only.

Glossary

Term	Meaning
BillableEvent	A catalog entry declaring a single chargeable event template. Carries trigger metadata, applicability, eligibility filters, optional state callback, signed-amount policy. Operator-scoped.
BillableEventCategory	Finance-reporting classification (LIFECYCLE_FEE, EARLY_TERMINATION_FEE, REFUND, etc.). Operator-scoped. Drives revenue-stream separation.
Trigger type	The mechanism that fires a BillableEvent: SAGA_INTENT (synchronous workflow billing call), LIFECYCLE_EVENT (async post-commit Kafka event), ADMIN_ACTION, CUSTOMER_PURCHASE, EXTERNAL_PAYMENT, SCHEDULED.

Term	Meaning
Saga intent code	Canonical intent from SUB-WF-FRAMEWORK-01: ACTIVATION_INTENT, PAUSE_INTENT, RESUME_INTENT, UPGRADE_INTENT, DOWNGRADE_INTENT, RELOCATION_INTENT, MIGRATION_INTENT, RESTRICTION_ADD_INTENT, RESTRICTION_REMOVE_INTENT, SUSPEND_NP_INTENT, TERMINATION_INTENT. A workflow's bil01-emit-intent call carries one of these. BIL-01 resolves matching active BillableEvents by (operatorCode, triggerType=SAGA_INTENT, triggerIntentCode) + filter rules.
State callback	An optional invocation of SUB-LM-01 by BIL-01 after a successful charge, used by events that drive subscription transitions (e.g., RECONNECTION_FEE_AFTER_DUNNING calls back to flip status SUSPENDED_NP → ACTIVE).
Amount sign policy	POSITIVE_ONLY (charge — increments customer balance owed), NEGATIVE_ONLY (credit — decrements / refunds), or SIGNED (runtime-computed signed amount, used for proration deltas where direction depends on source vs target Package price).
Filter rule	An optional Drools rule (in the billing-rules KJAR) that accepts/rejects a BillableEvent match based on the workflow's intentContext (e.g., reject EARLY_TERMINATION_FEE if the subscription is past its minimum-term).

1. What this DD owns

1. The billable_event_category table.
2. The billable_event table.
3. Rules governing trigger types, applicability, signed-amount semantics, state callbacks, eligibility filters, and operator scoping.
4. The CRUD API for catalog management (admin-only).
5. Kafka catalog-administration events (created/updated/activated/retired).

What this DD does NOT own

- **Charging logic** — BIL-01 (the engine that reads this catalog at runtime).
- **Service catalog** — PLM-CFG-01 (service_refs[] entries on BillableEvents resolve to ONE_SHOT Services there).
- **Currency derivation** — Services carry currency; BillableEvents inherit from referenced Services (R-B-4).
- **Tax computation** — Services carry default tax-group refs; BIL-01 applies them per PLM-CFG-02.
- **Customer wallet / payment** — handled downstream by BIL-01 + BIL-01-PAY-01.
- **State transitions** — SUB-LM-01 owns the actual transition mechanism; this DD only declares transition codes BillableEvents may invoke via callback.

Operator scoping

Every BillableEvent and Category is scoped to one operator. KE WIK seeds one catalog; WUG/WTZ/YASSN seed their own. Each operator's deployment adds, retires, or modifies its own catalog entries without affecting other operators. Cross-operator references are rejected at create/update time.

2. Schemas

2.1 billable_event_category

Finance reporting classification, operator-scoped.

```
CREATE TABLE billable_event_category (
  id                TEXT PRIMARY KEY,
  operator_code     TEXT NOT NULL,
  code              TEXT NOT NULL,
  description        TEXT NOT NULL,
  gl_account_hint   TEXT NULL,
  is_credit          BOOLEAN NOT NULL DEFAULT FALSE,
  display_order     INTEGER NOT NULL DEFAULT 0,
  status            TEXT NOT NULL,
  created_at        TIMESTAMPTZ NOT NULL DEFAULT NOW(),
```

```

        created_by          TEXT NOT NULL,
        updated_at          TIMESTAMPTZ NULL,
        updated_by          TEXT NULL,
        retired_at          TIMESTAMPTZ NULL,
        UNIQUE (operator_code, code) WHERE retired_at IS NULL,
        CHECK (status IN ('DRAFT', 'ACTIVE', 'RETIRED'))
    );

```

Column	Notes
operator_code	WIK / WUG / WTZ / YASSN
code	Uppercase snake-case. Unique per operator.
is_credit	true for refund / credit-issuing categories (REFUND, PROMOTIONAL_CREDIT, SLA_COMPENSATION, BUYBACK_CREDIT). Charges that ALSO have credit-style BillableEvents (e.g., proration delta that may go either way) use is_credit=false on the category and amount_sign_policy=SIGNED on the event.
gl_account_hint	Finance reconciliation hint. Advisory; not enforced by BIL-01.

Sample data (KE seed):

id	operator_code	code	description	is_credit
bec_01HX_LCY_WIK	WIK	LIFECYCLE_FEE	Subscription lifecycle fees (pause, reactivation, package change, address move)	false
bec_01HX_ETF_WIK	WIK	EARLY_TERMINATION_FEE	Early-termination fees	false
bec_01HX_INS_WIK	WIK	INSTALLATION_FEE	Installation and setup fees	false
bec_01HX_RCN_WIK	WIK	RECONNECTION_FEE	Fees for resuming service after non-payment suspension	false
bec_01HX_PRO_WIK	WIK	PRORATION	Proration deltas — partial-cycle charges or credits for cycle-anchor changes	false
bec_01HX_VAS_WIK	WIK	VAS	Value-added services (PPV, on-demand content, premium support)	false
bec_01HX_RGL_WIK	WIK	REGULATORY	Government-mandated regulatory fees and levies	false
bec_01HX_REF_WIK	WIK	REFUND	Customer refunds (cash, service credit, deposit return)	true
bec_01HX_SLA_WIK	WIK	SLA_COMPENSATION	SLA-driven service credits issued to customers	true
bec_01HX_BBK_WIK	WIK	BUYBACK_CREDIT	Hardware buyback credits issued to customers	true
bec_01HX_PRM_WIK	WIK	PROMOTIONAL_CREDIT	One-shot promotional or welcome credits	true

2.2 billable_event

The catalog of chargeable events, operator-scoped.

```

CREATE TABLE billable_event (
    id                      TEXT PRIMARY KEY,
    operator_code          TEXT NOT NULL,
    code                   TEXT NOT NULL,
    description             TEXT NOT NULL,
    category_code          TEXT NOT NULL,
    service_refs           JSONB NOT NULL,
    currency               TEXT NOT NULL,
    applicability           TEXT NOT NULL,
    amount_sign_policy     TEXT NOT NULL DEFAULT 'POSITIVE_ONLY',
    pay_first_required     BOOLEAN NOT NULL DEFAULT TRUE,
    trigger_type           TEXT NOT NULL,
    trigger_intent_code    TEXT NULL,
    trigger_event_type     TEXT NULL,
    trigger_filter_drl     TEXT NULL,
    trigger_schedule       TEXT NULL,
    state_callback         JSONB NULL,
    eligibility_franchise_refs JSONB NULL,
    eligibility_package_refs JSONB NULL,
    eligibility_segment_refs JSONB NULL,

```

```

display_order          INTEGER NOT NULL DEFAULT 100,
status                 TEXT NOT NULL,
notes                  TEXT NULL,
created_at             TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by             TEXT NOT NULL,
updated_at            TIMESTAMPTZ NULL,
updated_by            TEXT NULL,
retired_at            TIMESTAMPTZ NULL,
UNIQUE (operator_code, code) WHERE retired_at IS NULL,
CHECK (status IN ('DRAFT', 'ACTIVE', 'RETIRED')),
CHECK (amount_sign_policy IN ('POSITIVE_ONLY', 'NEGATIVE_ONLY', 'SIGNED')),
CHECK (applicability IN ('PREPAID_ONLY', 'POSTPAID_ONLY', 'ANY')),
CHECK (trigger_type IN ('SAGA_INTENT', 'LIFECYCLE_EVENT', 'ADMIN_ACTION', 'CUSTOMER_PURCHASE', 'EXTERNAL_PAYMENT', 'SCHEDULED'));

```

Column	Notes
operator_code	WIK / WUG / WTZ / YASSN
code	Uppercase snake-case business identifier. Unique per operator. Convention: <TRIGGER_VERB>_<NOUN>_<VARIANT> (e.g., RELOCATION_FEE_CROSS_FRANCHISE).
category_code	FK to billable_event_category(operator_code, code). Same-operator required.
service_refs	JSONB array of ONE_SHOT Service code refs from PLM-CFG-01 (same operator). All referenced Services share one currency.
currency	Derived from Service refs at create/update; immutable thereafter.
applicability	Filter on subscription billing mode: PREPAID_ONLY, POSTPAID_ONLY, ANY.
amount_sign_policy	POSITIVE_ONLY — BIL-01 always charges (e.g., RELOCATION_FEE); NEGATIVE_ONLY — BIL-01 always credits (e.g., DOWNGRADE_PRORATE_CREDIT, SLA_COMPENSATION); SIGNED — BIL-01 (or its filter rule) computes a signed amount at runtime (e.g., MIGRATION_PRORATE_DELTA may yield + or – depending on source vs target Package price).
pay_first_required	When true, the workflow waits for BIL-01 confirmation before proceeding past the billing step. Required true for events with state_callback.
trigger_type	See §3 — SAGA_INTENT / LIFECYCLE_EVENT / ADMIN_ACTION / CUSTOMER_PURCHASE / EXTERNAL_PAYMENT / SCHEDULED.
trigger_intent_code	Required when trigger_type = SAGA_INTENT. One of the canonical intents (§3.1).
trigger_event_type	Required when trigger_type = LIFECYCLE_EVENT. Kafka event name from SUB-LM-01 or other publishers.
trigger_filter_drl	Optional. Rule name in billing-rules KJAR. Provides additional filtering beyond intent-code match.
trigger_schedule	Required when trigger_type = SCHEDULED. Cron expression.
state_callback	Optional JSONB {transitionCode, targetStatus, callbackMetadata?} invoking SUB-LM-01 after successful charge.
eligibility_franchise_refs	Optional. Limits the event to subscriptions whose HomePass is in one of the listed Franchises. NULL or empty = all Franchises.
eligibility_package_refs	Optional. Limits to subscriptions of specific Packages.
eligibility_segment_refs	Optional, reserved for CVM segments.
display_order	Resolution order when multiple BillableEvents match the same trigger.
status	DRAFT / ACTIVE / RETIRED.

Indexes:

- idx_billable_event_code UNIQUE on (operator_code, code)
- idx_billable_event_trigger_intent partial on (operator_code, trigger_intent_code) WHERE status = 'ACTIVE' AND trigger_type = 'SAGA_INTENT'
- idx_billable_event_trigger_event partial on (operator_code, trigger_event_type) WHERE status = 'ACTIVE' AND trigger_type = 'LIFECYCLE_EVENT'
- idx_billable_event_category on (operator_code, category_code)
- idx_billable_event_applicability partial on (operator_code, applicability) WHERE status = 'ACTIVE'

Sample data (KE WIK seed — illustrative; deployment-specific values):

code	category	trigger_type	trigger_intent_code	trigger_filter_drl	amount_sign_policy	applicability	state_callback
INSTALLATION_FEE	INSTALLATION_FEE	SAGA_INTENT	ACTIVATION_INTENT	—	POSITIVE_ONLY	ANY	—
FIRST_CYCLE_PRORATE	PRORATION	SAGA_INTENT	ACTIVATION_INTENT	cycleAnchorOverflowApplied == PRORATE_PARTIAL_CYCLE	POSITIVE_ONLY	ANY	—
PAUSE_FEE_VACATION	LIFECYCLE_FEE	SAGA_INTENT	PAUSE_INTENT	pauseReasonCode == CUSTOMER_VACATION	POSITIVE_ONLY	ANY	—
RECONNECTION_FEE	RECONNECTION_FEE	SAGA_INTENT	RESUME_INTENT	resumeTrigger == CUSTOMER_REQUESTED_RESUME	POSITIVE_ONLY	ANY	—
UPGRADE_PRORATE_DELTA	PRORATION	SAGA_INTENT	UPGRADE_INTENT	cycleAnchorPolicy == PRESERVE	POSITIVE_ONLY	ANY	—
DOWNGRADE_PRORATE_CREDIT	PRORATION	SAGA_INTENT	DOWNGRADE_INTENT	cycleAnchorPolicy == PRESERVE AND effectiveTiming == IMMEDIATE	NEGATIVE_ONLY	ANY	—
OLD_CYCLE_CREDIT	PRORATION	SAGA_INTENT	UPGRADE_INTENT OR DOWNGRADE_INTENT OR MIGRATION_INTENT	cycleAnchorPolicy == RESET_TO_OPERATION_DATE	NEGATIVE_ONLY	ANY	—
FIRST_CYCLE_CHARGE	PRORATION	SAGA_INTENT	UPGRADE_INTENT OR DOWNGRADE_INTENT OR MIGRATION_INTENT	cycleAnchorPolicy == RESET_TO_OPERATION_DATE	POSITIVE_ONLY	ANY	—
RELOCATION_FEE_SAME_FRANCHISE	LIFECYCLE_FEE	SAGA_INTENT	RELOCATION_INTENT	sameFranchise == true	POSITIVE_ONLY	ANY	—
RELOCATION_FEE_CROSS_FRANCHISE	LIFECYCLE_FEE	SAGA_INTENT	RELOCATION_INTENT	sameFranchise == false	POSITIVE_ONLY	ANY	—
RELOCATION_FEE_NEW_INSTALL	LIFECYCLE_FEE	SAGA_INTENT	RELOCATION_INTENT	equipmentPath == NEW_INSTALL	POSITIVE_ONLY	ANY	—
MIGRATION_FEE	LIFECYCLE_FEE	SAGA_INTENT	MIGRATION_INTENT	migrationTrigger != ADMIN_Tech_MIGRATION	POSITIVE_ONLY	ANY	—
MIGRATION_PRORATE_DELTA	PRORATION	SAGA_INTENT	MIGRATION_INTENT	cycleAnchorPolicy == PRESERVE	SIGNED	ANY	—
RESTRICTION_LIFT_FEE_OUTGOING_VOICE	LIFECYCLE_FEE	SAGA_INTENT	RESTRICTION_REMOVE_INTENT	restrictionCode == OUTGOING_VOICE_BARRED	POSITIVE_ONLY	ANY	—
EARLY_TERMINATION_FEE	EARLY_TERMINATION_FEE	SAGA_INTENT	TERMINATION_INTENT	coolingOffApplicable == false AND daysActiveInContract < contractMinimumTermMonths * 30	POSITIVE_ONLY	ANY	—
PREPAID_REFUND	REFUND	SAGA_INTENT	TERMINATION_INTENT	prepaidWalletBalance > 0	NEGATIVE_ONLY	PREPAID_ONLY	—

code	category	trigger_type	trigger_intent_code	trigger_filter_drl	amount_sign_policy	applicability	state_callback
EQUIPMENT_REPLACEMENT_FEE	LIFECYCLE_FEE	SAGA_INTENT	RELOCATION_INTENT OR MIGRATION_INTENT	equipmentPath == NEW_INSTALL AND replacementReason != null	POSITIVE_ONLY	ANY	—
EQUIPMENT_BUYBACK_ONT	BUYBACK_CREDIT	ADMIN_ACTION	—	—	NEGATIVE_ONLY	ANY	{transitionCode: "EQUIPMENT_UNBIND", targetStatus: "ACTIVE"}
PPV_MOVIE	VAS	CUSTOMER_PURCHASE	—	—	POSITIVE_ONLY	ANY	—
VAT_SURCHARGE	REGULATORY	SCHEDULED	—	—	POSITIVE_ONLY	ANY	—
SLA_COMPENSATION_DOWNTIME	SLA_COMPENSATION	ADMIN_ACTION	—	—	NEGATIVE_ONLY	ANY	—
RECONNECTION_FEE_AFTER_DUNNING	RECONNECTION_FEE	EXTERNAL_PAYMENT	—	—	POSITIVE_ONLY	POSTPAID_ONLY	{transitionCode: "RECONNECT_AFTER_FEE", targetStatus: "ACTIVE"}

(All rows are operator_code = WIK. WUG/WTZ/YASSN seed their own catalogs with operator-appropriate codes, fees, eligibility filters, and trigger rules. The KE seed above is illustrative — actual fee amounts and applicability rules come from the workshop deep-dives for each operator.)

service_refs JSONB shape:

```
["svc_relocation_cross_franchise_01HX"]
```

state_callback JSONB shape:

```
{"transitionCode": "RECONNECT_AFTER_FEE", "targetStatus": "ACTIVE", "callbackMetadata": {"reason": "DUNNING_RESOLVED"}}
```

trigger_filter_drl references a rule name in the billing-rules KJAR (operator-scoped package). The rule receives the workflow's intentContext as facts and votes accept/reject. Example rule signature:

```
rule "filter_RelocationFeeSameFranchise"
when
  $ctx: IntentContextFact(sameFranchise == true)
then
  $ctx.acceptBillableEvent("RELOCATION_FEE_SAME_FRANCHISE");
end
```

3. Rules

3.1 BillableEvent integrity

Rule	Description
R-BIL-CFG-01-B-1	BillableEvent code is unique per (operator_code, code) excluding retired. Convention: uppercase snake-case, max 64 chars.
R-BIL-CFG-01-B-2	code is immutable once any charge has been booked under the event. The code is the audit-trail key; renaming would break historical ledger entries.
R-BIL-CFG-01-B-3	A BillableEvent MUST reference at least one Service in service_refs[]. Each Service ref MUST resolve to an ACTIVE Service in PLM-CFG-01 within the same operator, with consumption_model = ONE_SHOT. Continuous Services cannot be referenced — they belong to Packages, not BillableEvents.
R-BIL-CFG-01-B-4	currency is derived from Service refs at create/update time: all referenced Services share one currency, which becomes the BillableEvent's currency. Mixed-currency Service refs are rejected with 400 MIXED_CURRENCY_REFS. Currency is immutable thereafter.
R-BIL-CFG-01-B-5	applicability declares which subscription billing modes apply. Same shape as PLM-CFG-03's wallet applicability. BIL-01 filters at charge time: a PREPAID_ONLY event triggered against a postpaid subscription is skipped.

Rule	Description
R-BIL-CFG-01-B-6	pay_first_required MUST be true for any BillableEvent with a state_callback. The callback fires ONLY after successful charge confirmation. Setting pay_first_required = false on an event with a state_callback returns 400 PAY_FIRST_REQUIRED_FOR_STATE_CALLBACK.
R-BIL-CFG-01-B-7	category_code MUST reference an ACTIVE row in billable_event_category of the same operator. Cross-operator category reference returns 400 CATEGORY_OPERATOR_MISMATCH.
R-BIL-CFG-01-B-8	An ACTIVE BillableEvent cannot retire while any subscription has an in-flight workflow that has resolved this event as a match. Admin must let the workflow complete (or revert) before retirement.

3.2 Amount sign policy

Rule	Description
R-BIL-CFG-01-AS-1	amount_sign_policy ∈ {POSITIVE_ONLY, NEGATIVE_ONLY, SIGNED}.
R-BIL-CFG-01-AS-2	POSITIVE_ONLY — BIL-01 emits a charge with positive amount. Used for fees: RELOCATION_FEE, MIGRATION_FEE, INSTALLATION_FEE, EQUIPMENT_REPLACEMENT_FEE, etc. The category's is_credit is typically false.
R-BIL-CFG-01-AS-3	NEGATIVE_ONLY — BIL-01 emits a credit with negative amount. Used for refunds and committed credits: DOWNGRADE_PRORATE_CREDIT, OLD_CYCLE_CREDIT, PREPAID_REFUND, SLA_COMPENSATION_DOWNTIME, EQUIPMENT_BUYBACK_ONT. Category's is_credit is typically true.
R-BIL-CFG-01-AS-4	SIGNED — BIL-01's pricing rule (in billing-rules KJAR) computes the sign at runtime. Used for events where the direction depends on context, primarily MIGRATION_PRORATE_DELTA: target Package may be priced higher than source (charge) or lower (credit). The category.is_credit should be false; signing is per-call. The runtime rule MUST produce a signed amount; ambiguous outcomes return error.
R-BIL-CFG-01-AS-5	The sign policy is immutable once ACTIVE. Changing requires retire + re-create.

3.3 Trigger taxonomy

Rule	Description
R-BIL-CFG-01-T-1	trigger_type ∈ {SAGA_INTENT, LIFECYCLE_EVENT, ADMIN_ACTION, CUSTOMER_PURCHASE, EXTERNAL_PAYMENT, SCHEDULED}.
R-BIL-CFG-01-T-2	SAGA_INTENT — fires synchronously during a workflow's bil01-emit-intent call. Requires trigger_intent_code matching one of the canonical intents (§3.1 glossary). BIL-01 resolves matching BillableEvents by (operatorCode, trigger_type=SAGA_INTENT, trigger_intent_code) + filter rule, charges them in display_order then code, returns the aggregated line items to the workflow.
R-BIL-CFG-01-T-3	LIFECYCLE_EVENT — fires asynchronously on a Kafka event from SUB-LM-01 or other publishers (e.g., SubscriptionPaused, SubscriptionTerminated). Requires trigger_event_type. Used for post-commit reactive billing: notifications-with-billing, deferred ancillary fees. MUST NOT carry state_callback (the lifecycle has already committed).
R-BIL-CFG-01-T-4	trigger_filter_drl — optional Drools rule name for additional matching. Rule receives the workflow's intentContext (or event payload for LIFECYCLE_EVENT) and votes accept/reject. Storing logic as a rule rather than inline JSONB keeps the rule engine consistent across decision points.
R-BIL-CFG-01-T-5	ADMIN_ACTION / CUSTOMER_PURCHASE / EXTERNAL_PAYMENT — no intent code or event type required. Invoked by direct call to BIL-01's charging endpoint with the BillableEvent's code as the parameter.
R-BIL-CFG-01-T-6	SCHEDULED — requires trigger_schedule as a cron expression. A BIL-01 background job evaluates the schedule per operator and fires the event against each eligible customer.
R-BIL-CFG-01-T-7	Multiple BillableEvents may share the same trigger_intent_code (different filter rules). BIL-01 charges them in display_order then code order. Same for shared trigger_event_type on LIFECYCLE_EVENT triggers.
R-BIL-CFG-01-T-8	trigger_type is immutable once ACTIVE. Switching trigger model requires retire + re-create.

3.4 State callbacks

Rule	Description
R-BIL-CFG-01-SC-1	state_callback is OPTIONAL JSONB {transitionCode, targetStatus, callbackMetadata?}. The callback invokes SUB-LM-01's transition mechanism after successful charge. The callback is named by transition code, NOT by URL — SUB-LM-01 owns the mapping.
R-BIL-CFG-01-SC-2	transitionCode MUST be one of the registered transition codes published by SUB-LM-01. Validated against a known list. New codes require a coordinated update with SUB-LM-01.
R-BIL-CFG-01-SC-3	When a state callback is declared, pay_first_required MUST be true. The callback fires only on successful charge. On charge failure, the callback is NOT fired and the BillableEvent outcome event carries stateCallbackInvoked = false.
R-BIL-CFG-01-SC-4	callbackMetadata (optional) is carried verbatim to SUB-LM-01. Used for custom transition reasons, actor identity, channel hints.
R-BIL-CFG-01-SC-5	If the callback invocation fails (SUB-LM-01 unreachable, transition rejected), the charge is NOT rolled back. The BillableEvent emits BillableEventStateCallbackFailed for alerting; admin intervention reconciles. Money in the till is prioritized over state consistency; the rollback alternative is more complex than the recovery scenario warrants.

3.5 Eligibility filters

Rule	Description
R-BIL-CFG-01-EL-1	eligibility_franchise_refs[] (optional) restricts the BillableEvent to subscriptions whose HomePass is in one of the listed Franchises. Empty / NULL = all. Allows regional pricing variations within an operator.
R-BIL-CFG-01-EL-2	eligibility_package_refs[] (optional) restricts to specific Packages. Allows tier-specific fees.
R-BIL-CFG-01-EL-3	eligibility_segment_refs[] reserved for CVM customer segments (future). Not enforced in v1.0.

4. REST API

4.1 Create BillableEvent

POST /api/billing/billable-events HTTP/1.1
Content-Type: application/json

```
{
  "operatorCode":      "WIK",
  "code":              "MIGRATION_FEE",
  "description":       "Standard customer-initiated migration fee",
  "categoryCode":      "LIFECYCLE_FEE",
  "serviceRefs":       ["svc_migration_standard_01HX"],
  "applicability":     "ANY",
  "amountSignPolicy":  "POSITIVE_ONLY",
  "payFirstRequired":  true,
  "triggerType":       "SAGA_INTENT",
  "triggerIntentCode": "MIGRATION_INTENT",
  "triggerFilterDrl":  "filter_MigrationFeeStandard",
  "displayOrder":      100,
  "status":            "DRAFT",
  "createdBy":         "admin-wik-001"
}
```

→ 201 Created

```
{
  "id":                "be_01HX_MIGRATION_FEE_WIK",
  "operatorCode":      "WIK",
  "code":              "MIGRATION_FEE",
  "currency":          "KES",
  "status":            "DRAFT",
  "createdAt":         "2026-05-25T15:00:00+03:00"
}
```

4.2 Activate BillableEvent (DRAFT → ACTIVE)

POST /api/billing/billable-events/{id}/activate HTTP/1.1
Content-Type: application/json

```
{"activatedBy": "admin-wik-001"}
```

```
→ 200 OK
{"id": "be_01HX_MIGRATION_FEE_WIK", "status": "ACTIVE", "activatedAt": "2026-05-25T15:01:00+03:00"}
```

GET /api/billing/billable-events/{id} — read. GET /api/billing/billable-events?operatorCode=WIK&triggerType=SAGA_INTENT&triggerIntentCode=RELOCATION_INTENT&status=ACTIVE — filter (BIL-01's hot lookup). PATCH /api/billing/billable-events/{id} — update mutable fields (description, display_order, filter rule, eligibility filters, status). Rejects mutation of code, currency, triggerType, amountSignPolicy. POST /api/billing/billable-events/{id}/retire — soft retire (ACTIVE → RETIRED). Subject to R-B-8.

4.3 Errors

```
→ 400 Bad Request
{
  "code": "MIXED_CURRENCY_REFS",
  "guidance": "All Service refs must share one currency. Found mixed: ['KES', 'USD'].\"
}

→ 400 Bad Request
{
  "code": "PAY_FIRST_REQUIRED_FOR_STATE_CALLBACK",
  "guidance": "Events with a state_callback must have payFirstRequired=true. State callbacks fire only on successful charge\"
}

→ 409 Conflict
{
  "code": "BILLABLE_EVENT_IN_USE",
  "inFlightWorkflows": [\"subop_01HZ...\"],
  "guidance": \"Cannot retire while in-flight workflows have resolved this event as a match. Wait for workflow completion\"
}
```

5. Kafka events

Catalog administration events (this DD's events). Runtime charging events are owned by BIL-01.

Event	Topic	Fires on
BillableEventCreated	sophix.billing.billableevent.created	New row (DRAFT)
BillableEventUpdated	sophix.billing.billableevent.updated	Mutable column change
BillableEventActivated	sophix.billing.billableevent.activated	DRAFT → ACTIVE (BIL-01's cache invalidation trigger)
BillableEventRetired	sophix.billing.billableevent.retired	ACTIVE → RETIRED
BillableEventCategoryCreated / Updated / Retired	sophix.billing.billableeventcategory.*	Sub-catalog row lifecycle

Sample BillableEventActivated:

```
{
  "eventType": "BillableEventActivated",
  "aggregateId": "be_01HX_MIGRATION_FEE_WIK",
  "timestamp": "2026-05-25T15:01:00+03:00",
  "actor": "admin-wik-001",
  "payload": {
    "operatorCode": "WIK",
    "code": "MIGRATION_FEE",
    "description": "Standard customer-initiated migration fee",
    "categoryCode": "LIFECYCLE_FEE",
    "serviceRefs": [\"svc_migration_standard_01HX\"],
    "currency": "KES",
    "applicability": "ANY",
    "amountSignPolicy": "POSITIVE_ONLY",
    "payFirstRequired": true,
    "triggerType": "SAGA_INTENT",
    "triggerIntentCode": "MIGRATION_INTENT",
    "triggerFilterDrl": "filter_MigrationFeeStandard",
    "stateCallback": null,
    "displayOrder": 100,
    "status": "ACTIVE"
  }
}
```

6. Cross-DD coordination

Module	Direction	Contract
BIL-01 (Charging Engine)	Primary consumer	Reads catalog at runtime to resolve which events to charge on bil01-emit-intent calls. Subscribes to admin events for cache invalidation.
SUB-WF-* workflow DDs	Indirect consumers via BIL-01	Workflows send bil01-emit-intent with intent codes from §3.1; BIL-01 resolves matches against this catalog.
SUB-LM-01	State-callback target	Publishes the registered transition codes that state_callback.transitionCode may reference.
PLM-CFG-01	Referenced	service_refs[] resolve to ONE_SHOT Services per operator.
PLM-CFG-02 (Tax)	Indirect	Services carry tax-group refs; BIL-01 applies them.
PLM-CFG-03 (Wallet)	Indirect	applicability filter mirrors wallet applicability semantics.
CVM / Finance / EM-04	Reporting consumers	Use category_code for revenue-stream separation.

6.1 Build order

This DD deploys after:

1. PLM-CFG-01 v1.0 (ONE_SHOT Services available for service_refs)
2. PLM-CFG-02 v1.0, PLM-CFG-03 v1.0 (tax + wallet refs)
3. SUB-LM-01 v1.0 (transition codes registered)

Consumers depending on v1.0:

1. BIL-01 v1.1 (the intent engine — reads catalog to charge per workflow intent)
2. SUB-WF-* workflow DDs that call bil01-emit-intent (indirectly via BIL-01)

6.2 Stub debt + open coordination items

- **WUG / WTZ / YASSN catalog seeds** — pending operator deep dives.
- **Billing-rules KJAR rule names** — trigger_filter_drl references rules in the per-operator KJAR (billing-rules-<operator>). Initial rule names listed in §2.2 sample data are illustrative; actual rule names defined when BIL-01 v1.1 ships.

B — Current TZ

Status: To be filled in after codebase cartography review.

A BillableEvent-shaped catalog likely exists in the TZ codebase (the v0.8 baseline references workshop observation). Specific structural details — whether the existing catalog has operator scoping, whether signed-amount semantics are present, whether cycle-anchor RESET events are catalog rows or hardcoded — require source-code review to document accurately.